

AMPLIFIED: A Modular Toolchain to Benchmark Robotic Planning and Execution Tools

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Abstract

The wide catalogue of planners enables mission-planning solutions to be tailored to assumptions made on the environment and user needs. However, in robotics, the execution of plans and policies created by these planners is not always straightforward, because autonomous robots deployed in open and dynamic environments undergo fluctuating environmental conditions, partial information, and uncertain execution times. On top of bare engineering issues, planner integration and usability requires reliable and verifiable execution frameworks.

We propose AMPLIFIED, a complete toolchain to plan and execute in complex environments, with the goal to assess the quality of the integrated solution. AMPLIFIED is modular, thereby allowing the integration of community-made elements, from the decisional layer to low-level controllers. Building upon an antipoaching scenario, we illustrate how AMPLIFIED can be used to qualitatively evaluate different planners and the executability of their solutions. A video showcases such evaluation in simulated and real-life environments: <https://youtu.be/znwCHL5m1a4>.

Motivation

Autonomous systems often operate in dynamic, unpredictable environments for missions aimed at anti-poaching, search-and-rescue, or environmental monitoring, which all require robust planning and execution frameworks. AMPLIFIED addresses this need by providing a modular, ROS2 (Macenski et al. 2022) based toolchain that integrates deliberative planning, verifiable execution, and real-time adaptation. The toolchain enables qualitative evaluation and comparison of different planning and execution strategies, focusing on reliability, integrability, and usability. AMPLIFIED intends to accelerate the development of dependable planning for robotic systems, proposing a standardized simulation environment, integrating sensing and acting, as well as ensuring that plans are not only logically sound but also executable in real-world conditions. Additionally, the toolchain emphasizes formal verification and adaptive replanning, reducing mission failures and improving operational efficiency, while laying out the ground for more cognitive approaches to autonomous robots.

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AMPLIFIED Toolchain Description

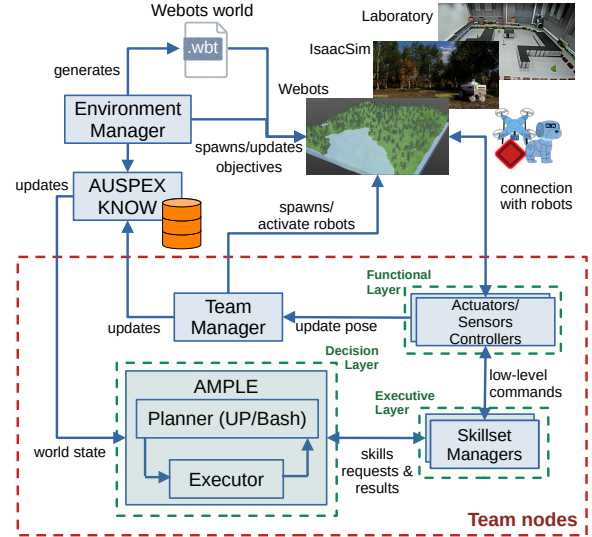


Figure 1: Decision, executive and the functional level of the AMPLIFIED simulation, planning and execution toolchain. Blue boxes represent the ROS2 nodes.

The blocks of AMPLIFIED, implemented using ROS2 and shown Figure 1, are:

- Environment and Team manager: modules solely dedicated as high-level interfaces with the simulation software (Webots, Isaac Sim) or real-life setting.
- AUSPEX-KNOW (Döschl, Sommer, and Kiam 2025): a world state knowledge database module that can store key/value pairs, as well as JSON structured data, updated and queried at any moment.
- For each team, a hierarchical framework to handle autonomous robot behaviors is deployed:
 - Decision Layer: uses AMPLIFIED (AnytiMe PLanning and Execution) (Chanel et al. 2019) to generate and adapt plans dynamically. It supports both sequential and proactive replanning strategies.
 - Execution Layer: a skill-based architecture (Albore et al. 2023) where robotic behaviors are defined

as modular, reusable skills (e.g., navigation, object manipulation). These skills can then be composed into complex behaviors and formally verified using Petri nets to ensure correctness (Pelletier, Lesire, and Godary-Dejean 2025).

- Functional Layer: connects to sensors and actuator controllers and simulated robotic components.

AMPLIFIED is designed to be modular, allowing users to swap components (e.g., planners, simulators) or extend functionality while maintaining compatibility through ROS2 interfaces. The default simulator, Webots, provides a realistic environment for multi-robot missions.

Demonstration Content

The demonstration will focus on an anti-poaching scenario where heterogeneous robot teams (UAVs and UGVs) collaborate to detect and neutralize poacher snares in a simulated forest in Webots, as shown in Figure 2. Robots must adapt to execution uncertainties, such as false detections or movement failures. To show the modularity of the toolchain, a video showcasing both simulated and real-life antipoaching missions is available: <https://youtu.be/znwcHL5mla4>.



Figure 2: AMPLIFIED will be showcased using Webots.

The AMPLIFIED toolchain will be demonstrated using four different planners implemented within the AMPLE module, in the decision layer:

- Fast Downward (Helmert 2006), a classical planner, called via Unified-Planning (Micheli et al. 2025).
- PANDA (Höller et al. 2021), a hierarchical planner.
- PR2 (Muise, McIlraith, and Beck 2024), a non-deterministic planner for handling uncertainties.
- Koala (Yousefi and Bercher 2024), a non-deterministic hierarchical planner.

The various elements of the toolchain will be presented individually, before a complete autonomous mission is showcased. The goal will be for the audience to appreciate how each block impacts the flow of the mission:

- Decisional Phase: The selected planner will generate an initial mission plan based on assumptions on the world state. Differences in integrability, replanning requirements, and plan efficiency will be highlighted.
- Execution Phase: Plans are translated into composite skills, verified via model-checking, and executed in Webots. Real-time adaptations of the skills generation and verification pipeline (e.g., replanning for failed actions) will be demonstrated.

- Knowledge Management: AUSPEX-KNOW updates the world state dynamically, which will be showcased in a high-level mission monitoring interface.

In order to summarize the results, qualitative and quantitative metrics have been prepared: integrability (API compatibility, output formats), usability, total planning time, re-planning frequency, and absence or presence of redundant actions in the solution plan. Results achieved for the used planners are available on Gitlab¹.

Acknowledgments

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¹https://gitlab.com/webots_plugins/amplified_bringup